

Introduction to Modern AI Agents - Assignment #2

Yair Zack (יאיר זק), ID 323127316

Andrei Nekliudov (אנדריי ניקליודוב), ID 345973044

Alan Arazi (אלן ארזי), ID 317430213

This document briefly presents 2 ideas for our course project, and drops the third unused one. To make this assignment self-contained, we left the content of the original assignment, and just added new content under the “Data Sources” subsection of both proposals. There are several sources that could be used to enable these kinds of applications. Our challenge would not be to find data, but to focus on a subset of it that makes it practical for a course project, without being too big, and defining a clear scope for the applications.

#1 Media & Creative Industries - Film & TV Production

The Problem / Pain Point

Movies and TV series are an important part of our life, but finding the “right” fit is not always simple, as we have different tastes and needs. ML recommendation systems such as Netflix tries to encourage content discovery, while adding explainability of why you might like something, but this is a very passive, static way. We believe recommendation systems and search engines can evolve to find matches that make more sense. In this idea we focus specifically on the pain-point of explicitly finding the “right” content to watch, given the user’s mood and intent. For example, consider the (real use-case by Alan) of “bring me a not-too-old Disney movie with no death and ideally involving animals for my 3 years old kid”.

The Idea (How the AI Agent solves it)

A platform like IMDB/Netflix has all the required information, starting from name, through the actors, metadata, plot description and even potentially every dialogue in the movie thanks to its subtitles. Yet, accessing this data is hard and querying even harder. An LLM equipped with access to such a database could answer complex-queries with semantic and logical filters such as the one mentioned above. Traditional RAG systems usually fail for such complex questions [1, 2], as relying on chunking and dense embeddings has both theoretical and practical limitations [3]. A potential problem here is that using a fixed K is limiting, as to properly iterate over all possible options, we might need to have a large-enough K. On the contrary, an Agentic solution could have a much smarter handling of retrieving, by decomposing the complex question into smaller ones, and accessing the relevant information. It could also use tools such as accessing database information. For example, some of the metadata such as they year and the movie studio could be retrieved with SQL-like filtering, which could be a much more efficient solution to focus on Disney movies.

Research Description

Efficient media content search and recommendation is a very important solution, and goes back to the “Netflix Prize” famous competition [4]. In modern days, people expect to have such agents much more developed; for IMDB, users complain about this function being very bad and requiring exact matches for years [5]. As a proof, we note that Netflix actively works on their recommendation systems [6], but it is very natural to expect that users could also actively search rather than passively get their recommendations, which is a different flow. Naturally, people could bypass IMDB or Netflix and directly query LLMs, but we show that using Gemini’s Thinking mode simply doesn’t get the desired outcome as expected - see the sequence of messages.



Reddit · r/imdb

10+ comments · 3 years ago

Why is the IMDb search function so bad?

Hi. If I search for something like "Will Penny 1968" in imdb.com - it returns nothing. In allmovie.com & rottentomatoes.com , it finds the Will Penny movie ...

14 answers · Top answer: I often just use Google to find the imdb page if the year is important. Though...

Why does **IMDb's search** function suck so bad? 4 answers 11 Aug 2015

Why is **IMDB** keep getting worse? 40 answers 6 Nov 2025

[More results from www.reddit.com](#)



Facebook · The Cinema Cafe

5 comments · 1 year ago

This is how bad the search on IMDb constantly is. You look ...

This is how **bad** the **search** on **IMDb** constantly is. You look for an exact phrase and that exact result comes as nr 7 or worse. Only **IMDb** knows why ... [Read more](#)



IMDb Community Forums

<https://community-imdb.sprinkl.com> › ... › IMDb.com

Search and "results do not appear" What stupid ...

19 Sept 2022 — In the Advanced **Search** box, there should be a link labeled "Enable adult titles/names in your **searches**." By clicking on "Enable", the film will ... [Read more](#)

Hi Alan

Where should we start?

I want to see a movie. bring me all, not-too-old (1990+) Disney movies with no deaths, that involve animals, for my 3 years old kid. Please be exhaustive - don't miss a single movie.

+ Tools

Thinking ▾



Disney Animal Movies for Toddlers

- **Ratatouille (2007):** While the legendary chef Gusteau is already deceased *before* the movie even begins, no characters die during the actual timeline of the movie. However, the film opens with an intense scene of an old woman firing a shotgun at a swarm of rats, and the heavy dialogue regarding culinary perfection and French restaurant politics will likely bore a toddler.

Ok, thanks, please bring the full list of all these movies in a bullet list.



Show thinking ▾

Here is the complete, exhaustive list of the movies discussed, broken out into individual bullet points for your convenience:

- Homeward Bound: The Incredible Journey (1993)
- 101 Dalmatians (1996)
- Homeward Bound II: Lost in San Francisco (1996)
- The Tigger Movie (2000)
- The Emperor's New Groove (2000)

are you completely sure you haven't forgotten about any movie?



Show thinking ▾

To be completely honest with you, when dealing with Disney's massive vault—especially when you factor in their direct-to-video sequels, midquels, and lesser-known live-action comedies—it is almost impossible to say "100% exhaustive" without a few more titles slipping through the cracks.

After digging even deeper into the archives to make absolutely sure nothing was missed, there are actually **several more titles** that strictly match your criteria (1990+, Disney, involves animals, zero deaths).

Here are the missing additions to complete your ultimate safe-watch list:

Additional Animated Gems

- **A Goofy Movie (1995)**
 - **Why it fits:** The entire cast consists of anthropomorphic animals (dogs, cats, etc.). It's a beautifully lighthearted, musical father-son road trip movie with zero deaths and great physical comedy that a toddler will find funny.

Data Sources

In order to work with the scope of the 50MB limitation, we can't really work with full scripts, as this will greatly reduce the scope of our work: 3,000 movie scripts are in the realm of 500MB. Therefore, for this POC we suggest using a reduced [source](#) of around 5K movies (13.9 MB), which offers rich metadata such as movie length, release year, or movie popularity, alongside free-text paragraphs that elaborate about the plot. This will allow us to create an interesting application that is able to both combine structured analysis ("bring movies from the past 10 years") alongside unstructured info ("bring movies with animals").

Nevertheless, we would like to emphasize that this is only a POC, and that we found a huge list of relevant, free and well-maintained data sources. Interestingly, the datasets contain both unstructured information (movie plots or transcripts), alongside rich metadata (ratings, year, duration, actors), and often they can be seen as a relational database which can serve multi-hop questions. While probably out of scope of this project, we would like to elaborate on these sources briefly, as our final project might eventually combine some of them.

Movies Databases

It is very common to have databases, which are stored as a relational datastructure which can be aggregated or queried. For example, a movie features several actors, and information about the actors can be stored in a different table. An example for this is [The Movies Dataset in Kaggle](#), which is built of several tables such as “keywords.csv”, “ratings.csv”, etc. While this source is almost 1GB, downsampling it to the relevant scope could work. It is worth noting that these corpuses are not structured per-se: they both contain long texts (movie descriptions) combined with metadata that can be filtered. Building solutions on top of such data can involve exciting opportunities of both classical RAG and SQL-like agentic operations over the metadata.

The Movies Dataset

Data Card Code (769) Discussion (43) Suggestions (3)

movies_metadata.csv (34.45 MB)

Detail

Compact

Column

10 of 24 columns

About this file

Contains Cast Crew Information for all movies in the movies_metadata.csv file.

id	imdb_id	original_language	original_title	overview
83%	45418 unique values	en 71% fr 5% Other (10759) 24%	43373 unique values	[null] 2% No overview found. 0% Other (44379) 98%
862	tt0114709	en	Toy Story	Led by Woody, Andy's toys live happily in his room until Andy's birthday brings Buzz Lightyear onto ...
8844	tt0113497	en	Jumanji	When siblings Judy and Peter discover an enchanted board game that opens the door to a magical world...
15602	tt0113228	en	Grumpier Old Men	A family wedding reignites the ancient feud between next-door neighbors and fishing buddies John and...

Data Explorer

Version 7 (943.76 MB)

credits.csv

keywords.csv

links.csv

links_small.csv

movies_metadata.csv

ratings.csv

ratings_small.csv

Summary

7 files

43 columns

Importantly, there are several variations of such data, here are a few examples:

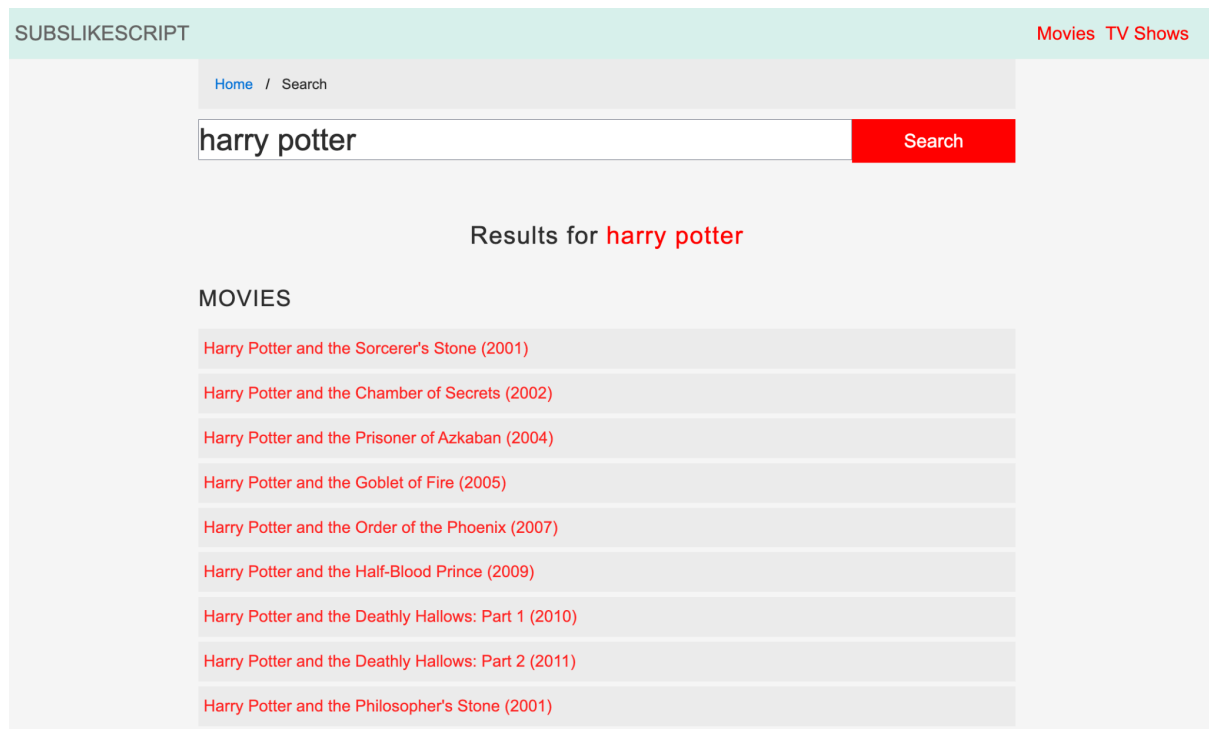
- <https://huggingface.co/datasets/AiresPucrs/tmdb-5000-movies> (13MB)
- <https://www.kaggle.com/datasets/ahmedosamamath/imdb-dataset> (312MB)
- <https://www.kaggle.com/datasets/raedaddala/imdb-movies-from-1960-to-2023> (241MB)

Movie Transcripts

Transcripts of movies are an incredibly rich source of long text that allow for complex querying. While there are websites that allow potentially scraping that content such as

[subslikescript.com](https://www.kaggle.com/datasets/fayaznoor10/movie-transcripts-59k), for the POC, we can rely on existing scrapes that are available on kaggle, for example the following ones:

- <https://www.kaggle.com/datasets/fayaznoor10/movie-transcripts-59k> (2.28GB)
- <https://www.kaggle.com/datasets/gufukuro/movie-scripts-corpus> (2.22GB)
- <https://www.kaggle.com/datasets/ismaeldwikat/movies-scripts> (246MB)
- https://huggingface.co/datasets/mocboch/movie_scripts (90MB)



Hybrid Source: Wikipedia / Wikidata

Another possible source is querying over Wikipedia pages. These movies have rich pages of data. If we take Frozen as an example, we can query over its Wikipedia [page](#), which is a complex object that provides access for both free-text and structured information. Also, it's common that these documents have a relatively shared structured, e.g. "Plot", allowing us to create agentic interfaces that query relevant areas of the pages, depending on the task. Wikipedia dumps are of course huge, but its downloading is documented (e.g. [Wikimedia](#)).

Frozen (2013 film)

🌐 99 languages ▾

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From Wikipedia, the free encyclopedia



Frozen is a 2013 American animated musical [fantasy film](#) produced by [Walt Disney Animation Studios](#),^[8] loosely based on [Hans Christian Andersen](#)'s 1844 fairy tale "[The Snow Queen](#)".^[1] Directed by [Chris Buck](#) and [Jennifer Lee](#) and written by Lee, it stars the voices of [Kristen Bell](#), [Idina Menzel](#), [Jonathan Groff](#), [Josh Gad](#), and [Santino Fontana](#). It follows [Anna](#), the princess of Arendelle, who sets off on a journey with the iceman [Kristoff](#), his reindeer [Sven](#), and the snowman [Olaf](#), to find her estranged sister [Elsa](#) after she accidentally traps their kingdom in eternal winter with her icy powers.

Frozen underwent several [story treatments](#) before it was commissioned in 2011. [Christophe Beck](#) was hired to compose the film's orchestral score, and [Robert Lopez](#) and [Kristen Anderson-Lopez](#) wrote the songs.

Following its world premiere at the [El Capitan Theatre](#) in Los Angeles on November 19, 2013, *Frozen* had its [general theatrical release](#) on November 27. It was praised for its visuals, screenplay, themes, music, and voice acting; some critics consider *Frozen* Disney's best animated film since the [Disney Renaissance](#). The film grossed over \$1.280 billion worldwide, becoming the [highest-grossing animated film](#) until the [remake of *The Lion King*](#) overtook this position in August 2019.^{[9][10]} It finished its theatrical run as the [highest-grossing film of 2013](#) and the [fifth-highest-grossing film of all time](#). The film's songs, characters, story, and appeal to a general audience led to it being dubbed a [pop culture](#) phenomenon.



Theatrical release poster

Directed by	Chris Buck Jennifer Lee
Screenplay by	Jennifer Lee
Story by	Chris Buck

References

- [1] Koshorek, Omri, et al. "Structured RAG for Answering Aggregative Questions." *arXiv preprint arXiv:2511.08505* (2025). (<https://arxiv.org/abs/2511.08505>)
- [2] Wolfson, Tomer, et al. "Monaco: More natural and complex questions for reasoning across dozens of documents." *Transactions of the Association for Computational Linguistics* 14 (2026): 23-46. (<https://direct.mit.edu/tacl/article/doi/10.1162/TACL.a.64/134745>)
- [3] Weller, Orion, et al. "On the theoretical limitations of embedding-based retrieval." *arXiv preprint arXiv:2508.21038* (2025). (<https://openreview.net/forum?id=k9CzlvzfaA>)
- [4] Netflix Prize, Wikipedia (https://en.wikipedia.org/wiki/Netflix_Prize)
- [5] <https://uxplanet.org/bad-ux-bites-10-imdbs-janky-search-is-why-you-need-design-leadership-5bff8c00660f>
- [6] <https://netflixtechblog.com/foundation-model-for-personalized-recommendation-1a0bd8e02d39>